

NexGen iMATCH

Forensic Integrity, Chain of Custody and Security Architecture

Field	Detail
System	NexGen iMATCH — forensic facial comparison and identification support
Document	Forensic Integrity and Security Brief
Revision	Revision 1, 5 August 2026
Audience	Forensic investigators, examiners, evaluating engineers, procurement reviewers
Status of system	Working system, verified locally. Not yet hosted, not yet independently validated.
Basis	Every control described here is implemented in code and verifiable in the repository. Items that are not implemented are marked as such rather than omitted.

The single sentence that governs this document

This system produces **investigative leads for human review**. It does not produce identifications. Every control described below exists to make that boundary structural rather than advisory — enforced by the code, not by the discipline of the operator.

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1. Purpose and scope

This brief sets out the forensic and security properties of NexGen iMATCH for evaluation by a forensic investigator or examiner. It is written to be checked rather than believed: each control names the mechanism that implements it, and each limitation is declared in the same document as the capability it qualifies.

The organising principle throughout is that **a forensic control which depends on an operator remembering it is not a control**. Where a rule could be enforced by the structure of the system, it has been — so that violating it is not merely against policy but not reachable through any code path.

This document deliberately includes the system's weaknesses. A capability statement that omits its own limitations invites an opposing expert to supply them instead, and the weakest unqualified claim in a document sets the credibility ceiling for everything else in it.

2. What the system is, and what it is not

2.1 Core functions

Function	What it does	What it outputs
1:1 verification	Compares two face images	A similarity score, a decision against a calibrated threshold, and a written explanation
1:N identification	Searches a probe face against an enrolled gallery	A ranked candidate list, each entry pending human adjudication
Batch comparison	Processes multiple images in one operation (three modes)	Per-item results with error isolation — one corrupt file does not fail the batch
Image enhancement	Improves a degraded CCTV image for visual examination	A separately stored, separately labelled analysis intermediate — never evidence
Case reporting	Compiles a case into a signed PDF report	Material examined, comparison plates, methodology, limitations, adjudication trail

2.2 What it is not

- **It is not an identification system.** No code path in the engine writes a confirmed match. Confirmation is an act performed by a named human examiner and recorded as such.
- **It is not a presentation-attack detector.** The liveness, deepfake and morphing screens are heuristics. None has been evaluated against ISO/IEC 30107-3. Each self-reports `certified: false` in its own output, and that label is carried through the API, the report and the user interface.
- **It is not an image-enhancement product for evidence.** Enhanced imagery exists for visual examination only and is structurally barred from the evidential path.
- **It is not independently validated.** Every accuracy figure in this document was produced by the team that built the system, on public academic datasets. This is stated as the single largest credibility gap in the programme.

3. Chain of custody — the seven controls

Chain of custody is not one feature; it is the conjunction of seven independent controls. Each is listed with the mechanism that implements it, so that any of them can be inspected directly.

#	Control	Mechanism	What it defeats
1	The original is never modified	Submitted bytes are written once to a content-addressed object store. Every downstream operation — detection, alignment, enhancement — works on a copy. Enhanced output is a separate object in a separate storage category, with its own hash.	Any allegation that the exhibit was altered in handling
2	Every file is identified by its own content	The filename of every stored image is its SHA-256 digest. Client-supplied filenames are never used, and paths are derived entirely from the digest.	Substitution, misfiling, and path-traversal attack in one control
3	Every record ties back to exact bytes	Database rows carry <code>image_sha256</code> and <code>probe_sha256</code> . A report, a search run and the file on disk can be reconciled by hash at any later date.	"Is this the same image the search was run on?"
4	Every consequential action is recorded	Enrolment, search, verification, enhancement, adjudication, export, login, key use and account changes all write an audit record.	Undocumented handling
5	The record cannot be silently edited	Audit records form a per-tenant hash chain, each entry hashing its own content, its explicit chain position and the previous entry's hash. Mirrored to JSONL on disk.	Retrospective alteration of the handling record
6	Every action names a person	Operator identity is resolved from a verified credential, never from the request body. Adjudications additionally record the examiner, the timestamp and free-text notes.	Anonymous or disputed attribution
7	Every search states its authority	<code>lawful_basis</code> is a mandatory field on searches and cases, stored verbatim in the audit chain.	"Why was this search performed?", asked months later

Storage layout, so it can be inspected

`storage/<tenant>/<probes | enrolments | enhanced>/ab/cd/abcd...ef.jpg`

The two-level fan-out exists because a single flat directory holding 10^5 files is painful on most filesystems. The tenant segment is the first isolation boundary; the digest is the identity of the exhibit.

4. Evidential separation of enhanced imagery

Super-resolution and face restoration synthesise detail from a learned prior. On a twenty-pixel face a large fraction of the restored output is the generator's prior rather than the subject. Presenting such an image as evidence, or matching against it without saying so, is the most serious failure mode available to a system of this kind.

4.1 The separation is enforced by type, not by policy

Type	Meaning	Admissible as evidence
OriginalImage	The bytes as received	Yes — and only this
RestoredImage	Deterministic operations only: deblocking, bounded deblur, interpolation. No synthesised detail.	No — analysis intermediate
ReconstructedImage	A learned prior contributed pixels	No — must be labelled a reconstruction and shown beside its original

The guard function that admits an image to the evidential path accepts `OriginalImage` and nothing else. An enhanced image cannot reach that path by mistake, because there is no conversion that would let it.

4.2 The supporting controls

- **Off by default.** The generative track is disabled unless explicitly enabled by configuration. Model weights are never downloaded implicitly and are checksum-verified when pinned.
- **Recorded when used.** Each enhancement writes an `EnhancementRun` row holding the original digest and path, the enhanced digest and path, the track (*restored* or *reconstructed*), the rule-set version, the device, and the timing.
- **Explainable, including what was not done.** The processing plan records the stages selected *and* the stages considered and skipped, each with its reason. An examiner override is recorded as a skip with its reason, not hidden.
- **Recognition runs A/B.** Where an enhanced probe is used for recognition, the original is the primary run and both searches record their source kind and the enhancement run identifier.
- **Bounded deblur.** The deblur correction is projected onto the measured passband of the image, so it cannot add energy above what the sensor actually recorded. This is tested, not asserted.

A deliberate failure, and why it is left in place

The report renderer's paired-enhanced-image function raises an error on purpose. An enhanced image may appear in a report *only* when labelled “AI-enhanced preview — not evidentiary, for visual reference only” **and** shown beside the unmodified original. A partial implementation could render the enhanced image without that pairing, which would breach the evidentiary rule. It therefore fails loudly rather than working incorrectly.

4.3 Measured, not assumed

Enhancement was evaluated on SCface — real mugshots against real CCTV capture at known distances — against a decision rule fixed before any run. Every enhancement arm was **rejected**: none beat the untransformed baseline, and none beat the plain-interpolation control. The operational conclusion is stated plainly: enhancement is for human visual examination; it is not a route to better matching.

5. Human authority over every conclusion

The engine ranks; it does not conclude. There is no code path that writes a confirmed adjudication. Confirmation and elimination are both performed by a person, and the record retains who, when, and on what note.

Stage	Actor	Recorded
Probe submitted	Operator	Operator id, lawful basis, purpose, probe digest, timestamp
Candidates returned	System	Rank, similarity, normalised score, gallery size, model and pack
Adjudication	Human examiner only	Examiner id, decision (confirmed / eliminated / pending), timestamp, examiner notes
Report issued	Examiner	Report digest, generating user, caveats rendered on every page

Enrolment additionally requires supervisor authority, because enrolment determines who the system is *capable* of finding. That is a different order of decision from running a search, and the role model reflects it.

6. Lawful basis, recorded and preserved

Every search carries a stated lawful basis, and that statement is written verbatim into the audit chain alongside the operator and the probe digest.

The claim, stated precisely

The system cannot judge whether a search is lawful. What it can do — and does — is ensure that somebody had to state a reason before the search would run, and that the statement is preserved in a record that cannot be silently edited afterwards.

7. Tamper-evident audit trail

7.1 Construction

- One chain per tenant. Each record's hash covers its own content, its explicit chain position, and the previous record's hash.
- Chain position is an explicit sequence column, **not** a timestamp ordering.
- A unique index on (tenant, sequence) turns concurrent appends into a loud insert failure rather than a silent fork of the chain.
- The chain is mirrored to a JSONL file on disk, so it survives compromise of the database alone.
- Any authenticated user may read their own tenant's trail, including entries about themselves.

Why ordering is not by timestamp — a defect found and fixed

created_at has millisecond resolution at best, so a burst of writes shares a tick. Ordering by time with a random-UUID tiebreaker made verification non-deterministic: the system could report tampering on an untouched log. For a forensic system that false positive is close to the worst possible defect, because it makes genuine detection indistinguishable from noise. The fix was an explicit sequence column with a uniqueness constraint.

7.2 What it proves, and what it does not

Statement	
Proves	No record has been altered or removed since it was written. Editing or deleting one breaks verification from that point onward, and a gap in the sequence is reported as a removed record.
Does not prove	Completeness. A party with database access could truncate the tail. Where that guarantee is required, the mirrored JSONL must be shipped to write-once storage. This is stated as a deployment requirement, not left to be discovered.

8. Cryptographic protection and isolation

8.1 Biometric templates

- A face is stored as a 512-dimension template, encrypted at rest with **AES-256-GCM**. The plaintext embedding is never written to the database.
- The key is a **per-tenant HKDF subkey**, with the tenant identifier bound in as additional authenticated data. A ciphertext moved between tenant rows *fails to decrypt* rather than silently matching under the wrong tenant.
- GCM is authenticated encryption: tampering raises an error rather than decrypting to plausible noise.
- **Templates never leave the server.** An embedding can be inverted into a recognisable approximation of the face it came from, so it is treated as equivalent to the photograph. The API returns template metadata only.
- Biometrics are irrevocable. A leaked template cannot be reissued like a password, which is why none of this is optional.

8.2 Tenant isolation

Isolation is **structural, not advisory**. Each tenant's vectors occupy a separate matrix, and the search function takes the tenant identifier as a required argument — so no code path exists that could compare a probe against another tenant's templates. A filter applied after a global search would be one forgotten predicate away from a cross-tenant biometric leak.

- The tenant always comes from the verified credential, never from a request body or header.
- Cross-tenant lookups return **404, not 403**: a distinguishable 403 would confirm that a given case or subject identifier exists.

8.3 Authentication and access control

Control	Implementation
Password storage	Argon2id at OWASP interactive parameters, with transparent rehash when cost parameters change
Login timing	Constant-time — a verification always runs, even with no matching user, so response timing cannot enumerate registered accounts
Error messages	Identical text for unknown user and wrong password
Sessions	Short-lived access tokens with separate refresh tokens; a refresh token presented where an access token is expected is rejected, so a long-lived credential cannot silently become a session
Account protection	E-mail verification, one-time-code checks with attempt and send-rate limits, failed-attempt lockout, and a session epoch that revokes all live sessions
API keys	32 bytes of CSPRNG output, stored as a SHA-256 hash, displayed once, never recoverable. Revocation deactivates rather than deletes, so audit references stay resolvable
Roles	investigator → supervisor → admin. Enrolment requires supervisor
Rate limits	120 requests per minute general; 30 per minute on search
Upload guard	15 MB ceiling, checked before and after base64 decoding; format identified from magic bytes rather than the declared content type

8.4 Transport and application security

- Explicit CORS origins — never a wildcard, which is both invalid and dangerous with credentialed requests.

- **Cache-Control:** no-store on every response — biometric findings must not sit in a shared cache. Plus X-Content-Type-Options, X-Frame-Options: DENY, Referrer-Policy: no-referrer, a correlation X-Request-ID, CSRF protection, and HSTS in production.
- Unhandled exceptions return a generic message plus the request id. Stack traces routinely leak schema, file paths and occasionally credentials.
- **Server-side URL import is refused (501).** Fetching a caller-supplied URL from inside the service is a server-side request forgery primitive that would reach internal addresses and cloud metadata endpoints using the server's own network position.
- Adversarial input handling is tested: 23 tests covering corrupt bytes, zero-byte files, extreme aspect ratios and non-face images. All return typed 4xx responses; none returns a 500.

9. Data lifecycle: retention and erasure

Data	Retention	Deletion behaviour
Probe images (searched)	Configurable window, default 90 days	Purged automatically once past the window
Enrolment images	Life of the subject record	Removed when the subject is deleted — not on a timer, because they belong to a record rather than to an event
Biometric templates	Life of the subject record	Removed outright on subject deletion
Gallery index entries	Life of the subject record	Removed outright on subject deletion
Search history	Retained	Deliberately retained — deleting it would destroy the audit trail. Candidate rows keep the subject identifier only
Audit records	Retained	Append-only by construction; deletion is detectable

Erasure is real. Deleting a subject removes templates, enrolment images and index entries outright — not a soft-delete flag. What survives is the record that a search occurred, which is itself a governance requirement rather than a residue of personal data.

10. CCTV-specific forensic analysis

Published restoration benchmarks do not contain the degradations that dominate real surveillance footage. Each difference below is measured, and each has a defined handler.

Condition	Why it matters forensically	Handler
Infrared (night) capture	Under IR illumination the image carries <i>no</i> colour information, and reflectances are infrared rather than visible. Every face restorer is trained on visible-light faces and will confidently produce plausible daylight skin tone — all of it invented, and it looks natural.	Detected from inter-channel correlation and near-zero saturation. Colour restoration is disabled and a mandatory report line is emitted: <i>“Infrared capture: the original contains no colour information. Any colour in the enhanced image is synthetic.”</i> This is a hard rule, not an advisory.
Inter-frame prediction (P/B frames)	A predicted frame's pixels are largely copied from a reference frame. Enhancing one can sharpen detail that belongs to <i>another instant</i> and then present it under this frame's timestamp.	Frame type is parsed from the bitstream and intra-coded frames are preferred. Non-intra frames remain usable but are labelled with their type and reference distance.
Interlacing	Comb artefacts on motion are treated by denoisers and super-resolution models as genuine high-frequency detail, and sharpened.	Detected by row-parity comb energy. Fields are separated rather than blended — they are two genuine temporal samples, and blending discards a real observation.
Overstated resolution	Recorders routinely upscale before storage. A “1080p” export can carry no real detail above 480p, and running super-resolution on it multiplies invention without adding information.	True effective resolution is measured from the image's own spectral band-limit and reported beside the stored pixel dimensions.
Unreliable degradation estimates	A wrong blur or compression estimate silently corrupts every downstream decision.	Blur, noise and compression are each estimated with a confidence measure, and every processing trigger is gated on that confidence. Compression-quality estimation is validated against known ground truth at 24 of 24 test conditions.

11. Evidence interpretation layer

A similarity score travels badly. Written into a report it loses the model that produced it, the population it was weighed against, the conditions on both sides, and the fact that it was never validated at the operating point being quoted. By the time it reaches a courtroom it has become “the system said 0.62”, which is not a scientific statement.

11.1 The reportable object

The evidence report object is built so that cannot happen. Provenance, calibration source, reference population, confidence interval and limitations are **required fields, not optional decoration**, and the renderers will not emit a number without them.

- **Calibrated likelihood ratio** with its reference population and confidence interval.
- **ENFSI verbal scale** (Guideline for Evaluative Reporting, 2015), symmetric about $LR = 1$. The verbal label is a communication aid for a fixed numeric range — never a substitute for the ratio, and always printed alongside it.
- **Discrimination metrics** — log-likelihood-ratio cost, pool-adjacent-violators calibration, Tippett plots.
- **Open-set handling** and random-match probability against a declared reference population.

The capacity guard

A reported likelihood ratio is checked against the identity information the observation can actually support. **A likelihood ratio larger than the evidence allows is the signature of a model prior leaking into a conclusion.** The guard flags and records rather than silently clamping, because the information bound is an expectation over comparisons rather than a hard per-case limit — and a silent clamp would itself be an undisclosed intervention.

Status. This layer is implemented and tested in the recognition engine and is used in the measurement and research path. It is *not yet wired into the product report*, which currently carries material examined, comparison plates, methodology, limitations, conclusion and the full adjudication trail. Completing that wiring is the next scheduled step and is recorded as outstanding rather than implied to be finished.

12. Report discipline

12.1 Photo-anthropometry is refused, deliberately

The obvious build is to detect landmarks, draw calipers on both faces, tabulate inter-landmark distances and title the section “why these images match”. That has not been done, and the reason is substantive rather than technical.

- Deriving identity conclusions from inter-landmark distances in 2D photographs has been explicitly rejected as an identification method by both FISWG and ENFSI.
- A photograph is a projection: any inter-landmark distance varies with yaw, pitch, camera-to-subject distance and focal length.
- Measurement variance for *the same person across two images* routinely exceeds the between-person variance of the same index. On degraded imagery, landmark placement error is itself large relative to the differences being reported.

Measurements are therefore printed as documented observations with error bars and an explicit pose caveat, never as grounds for the conclusion. Per-index differences are never combined into a composite score: they

are strongly correlated, and multiplying them manufactures evidence. **The reasoning is practical as well as principled** — if the report asserts that measurements demonstrate the match, an opposing expert cites that literature, and the calibrated likelihood ratio, which is the actually defensible number, is discredited alongside it.

12.2 Non-matches receive the identical structure

For an eliminated candidate the report highlights indices falling outside the same-source distribution, lists marks present in one image and absent in the other, and reports a likelihood ratio below one with its verbal band. “Why not” receives the same rigour as “why”. The symmetry is itself a credibility signal, and its absence is the first thing a reviewer notices.

12.3 Constraints on model-drafted prose

The report may optionally include model-drafted narrative. It is **disabled by default**, and where enabled it may write three descriptive sections only: an executive summary, a findings narrative, and an explanation of what the similarity figures mean.

Section	Who writes it	Reasoning
Executive summary	May be drafted	Descriptive only
Findings narrative	May be drafted	Descriptive only
What the figures mean	May be drafted	Descriptive only
Methodology	Template only	A paraphrased methodology can only introduce error about our own system
Limitations	Template only	A generated limitation can be a silently <i>softened</i> limitation
Conclusion	Template only	The conclusion <i>is</i> the decision; generating it would be the model deciding

Three guards run on any drafted text, and all three are tested:

Guard	Rule	The failure it blocks
Numeric whitelist	Every numeral in generated text must already appear in the evidence payload	A cosine similarity of 0.62 written up as “62% confidence” — exactly the sentence a reader would quote
Appearance vocabulary	Any sentence about a nose, jawline or hairline is rejected outright	The model receives numbers and never sees an image, so such a sentence is fabricated by construction
Assertion phrases	Language converting a similarity score into an identification is rejected, however it is hedged	Overstatement of what the comparison establishes

A failed generation retries once with the complaint fed back, then is abandoned and the report states that the narrative was withheld.

13. Measurement integrity

The controls in this section exist because the most likely source of a wrong forensic conclusion is not a compromised system — it is a measurement that was never correct.

- **One source of truth for thresholds.** Decision thresholds live in a single configuration object. They previously existed in four copies and had drifted, with the interface stating a rule the engine had stopped applying. Any model, threshold or fusion change now triggers a regression suite that fails loudly.
- **Withdrawn figures are published with their cause.** The measurement record carries a table of superseded numbers, each with its replacement and the defect that produced it — for example a rank-1 figure corrected from 37.43% to 32.93% after comparison pairs were found to have been built by sort order rather than by the official protocol.
- **Defects in our own measurement code are recorded, not repaired quietly.** Twelve were found during the current campaign, four of which had produced published numbers. Each is listed with symptom, cause and fix.
- **A test that looks like a lock and is not is treated as a defect.** One regression test was found whose docstring claimed a property its assertions did not check, and which imported the wrong module. It was rewritten and the episode recorded.

- **Decision rules are registered before the run.** Experiments carry pass and fail thresholds fixed in code beforehand. Where a result failed a criterion that had been set in advance, it was recorded as a failure rather than reinterpreted.
- **Harness validation against an external protocol.** The measurement harness reproduces published results on the official IJB-B and IJB-C protocols (95.44% and 96.91% true-accept at a false-accept rate of 1 in 10,000), which is what establishes that our figures are not artefacts of our own tooling.

14. Accuracy, stated by operating condition

The system's defining reporting rule is that accuracy is **never blended across operating conditions**. Clean-imagery and surveillance-imagery performance differ by tens of points and describe different real-world situations. A single averaged figure would not be technically false; it would be operationally misleading.

14.1 The three terms, and how they trade against each other

Term	Meaning	Who is harmed when it is high
False accept rate (FAR / FMR)	Two different people reported as a match	An innocent person is placed under investigation
False reject rate (FRR / FNMR)	The same person reported as different	A genuine lead is lost
True accept rate (TAR)	The same person correctly matched. $TAR = 100\% - FRR$	—

Raising the decision threshold lowers the false-accept rate and raises the false-reject rate. The two cannot be reduced together. Accuracy figures are therefore quoted as **true-accept rate at a fixed false-accept rate**, because a true-accept rate quoted alone can be made to reach 100% by accepting everything.

14.2 The deployed operating point, and the incident that set it

A real 1:1 comparison of two different people returned a similarity of 0.2405 and the system reported that the images supported the same-person proposition. This was not a defect: the pair fell between the then-deployed threshold and the target false-match point, and it is a direct instance of the false-match rate that had already been measured. The threshold was raised from 0.20 to 0.2871, and the cost was measured and accepted rather than assumed.

Group	FRR at 0.20	FRR at 0.2871	FAR at 0.20	FAR at 0.2871
All	3.30%	6.32%	1.19%	0.10%
Women	4.88%	8.45%	1.70%	0.16%
Men	2.21%	4.86%	0.68%	0.03%
Age 0–25	7.61%	14.75%	1.21%	0.10%
Age 26–40	3.24%	5.78%	1.26%	0.05%
Age 41–55	2.14%	3.91%	1.10%	0.15%
Age 56+	2.90%	6.12%	1.19%	0.09%

Measured on 40,098 pairs from AgeDB, with impostor pairs matched within subgroup — same gender, same age band — because cross-group impostors are trivially easy and would deflate every false-match rate. One global threshold was used rather than per-group thresholds, which would have concealed the very differential the audit exists to find.

Net effect: false matches fell approximately twelvefold, from about 1 in 84 comparisons to about 1 in 1,000. Missed genuine pairs roughly doubled. The rationale is that a missed lead costs a re-check, while a false match points investigators at an innocent person — so the two errors are not equivalent and the accuracy-maximising threshold was the wrong choice for forensic use.

14.3 Performance by operating condition

Condition	Corpus	True accept at 1-in-1,000 false accept
Clean, passport or custody quality	LFW	99.70%
Mixed quality, official protocol	IJB-C (at 1-in-10,000)	96.91%
Surveillance, approx. 32-pixel faces	TinyFace	20.59% — improved to 60.77% with the current backbone
Surveillance, approx. 25-pixel faces	QMUL-SurvFace	2.31% — improved to 22.35% after fine-tuning on real low-resolution data

The most important operational limitation, stated in full

On real cross-camera surveillance imagery the system **cannot reliably determine that a person is not in the gallery**. Tested against 1,844 genuine strangers, the mean top-ranked score for an enrolled person and for a stranger differed by 0.001 — with the stranger scoring marginally higher. No threshold separates them. The measured cause is an information deficit rather than a model deficiency: a 25-pixel face supplies roughly 2.9 bits of identity information against a gallery of that size demanding about 11.5. Training on real low-resolution imagery moved this metric for the first time, but it remains a starting point rather than a capability. **This is why human adjudication is not optional, and why the system will always return a ranked list even when the subject is absent from it.**

15. Declared limitations

Listed here in the same document as the capabilities they qualify, because a capability statement that omits them is not usable for evaluation.

#	Limitation
1	Not deployed. The system is verified locally. There is no hosted backend in the current configuration.
2	No independent validation. Every figure was produced by the same team and tooling that built the system, on public academic datasets. This is the largest credibility gap and nothing internal can close it.
3	Surveillance performance is weak, and stranger rejection on real cross-camera footage is effectively absent (§14.3).
4	Demographic differentials persist. Women are falsely rejected roughly 1.7 times as often as men; the under-25 band roughly 3.8 times as often as the 41–55 band. Raising the threshold relocated the errors without removing the gap. This must be re-measured on the deploying organisation's own population.
5	Threshold calibration is single-dataset. It is derived from AgeDB pairs and has not been validated against operational imagery. It is an operating point, not an accuracy guarantee.
6	Liveness, deepfake and morphing screens are heuristics, not certified detection, and a determined attacker defeats all three. Use them to prioritise examiner attention, never to conclude that media is authentic.
7	The evidence interpretation layer is not yet wired into the product report (§11).
8	Audit completeness is not cryptographically guaranteed without shipping the mirrored log to write-once storage (§7.2).
9	SQLite is the default database. It will not survive concurrent production load; PostgreSQL support is configured and tested but must be selected for deployment.
10	Development hardware bounds what can be attempted. All work to date was carried out on a single 6 GB laptop GPU. That ceiling is the reason several approaches were not pursued: the leading generative restorer requires 12–24 GB and was ruled out; full fine-tuning of the recognition backbone does not fit in memory at a usable batch size and was replaced with a low-rank adapter training roughly 1.2% of parameters. Available forensic corpora are also small and licence-restricted — SCface, for example, was used strictly as an evaluation corpus for the pre-registered comparison, never as training data. Both constraints bound the ceiling of the results reported here, and neither is a property of the method.

15.1 Required before production use

- Set and back up the JWT signing secret and the template encryption key. Without the latter, templates are stored unencrypted — the system logs this loudly, but it must not be allowed to happen.
- Calibrate thresholds on imagery representative of the intended casework.
- Move to PostgreSQL; terminate TLS at the ingress; restrict CORS origins.
- Complete legal, privacy and data-protection impact review for the jurisdiction.
- **Evaluate demographic performance differentials on your own population.** This is not a formality: a system deployed without measuring it will distribute its errors unevenly across the people it is used on, and nobody will notice.
- Define retention and deletion policy and set the probe retention window.
- Ship the audit JSONL to append-only storage.
- Establish examiner training and an adjudication standard.

16. How to verify each claim

Nothing in this document needs to be taken on trust. The following checks can be run by an evaluating engineer against the delivered system.

Claim	Check
Templates round-trip from encrypted storage	Enrol a subject, record the similarity, restart the service, search again — the score returns bit-identical
Audit chain is intact	Run the chain verification endpoint; then alter one record directly in the database and observe verification fail from that point onward
Tenant isolation is structural	Attempt a cross-tenant lookup with a valid credential — the response is 404, and no comparison is performed
Enhanced images cannot become evidence	Attempt to pass an enhanced image to the evidential path; the type guard rejects it
Accuracy figures are reproducible	Every published figure names the artifact file that produced it; the regression suite reproduces them across five datasets
Hostile input is handled	Submit corrupt bytes, zero-byte files, extreme aspect ratios and non-face images — all return typed 4xx responses
Thresholds have a single source	Change a threshold and observe the regression gate fail until the benchmark suite is re-run

Closing statement

This is a working forensic face-comparison system whose measurement foundation is stronger than its feature completeness. Its accuracy on clean imagery is at parity with the open-source state of the art because it *is* the open-source state of the art. Its distinguishing property is not accuracy: it is that the claims are measured, separated by operating condition, structurally constrained in what they may assert, and corrected in public when found wrong. It is not deployed, not independently validated, and not reliable on surveillance-quality imagery — and each of those is stated here rather than left to be discovered.